

RWANDA TVET BOARD
COMPETENCY-BASED ASSESSMENT
[SUMMATIVE ASSESSMENT]

Program: SOFTWARE DEVELOPMENT | **Level:** 4 | **Total Marks:** 100

Module: Backend Application Development Using Node JS | **Date:** 2026-05-26 | **Time:** 3 Hours

Name: _____ Index No: _____ Class: _____

SECTION A: TRUE / FALSE

Answer True (T) or False (F).

- 1 Understanding Middleware is necessary when developing RESTFUL APIs with Node JS. (2 marks)

Answer: **True** **False**

- 2 RESTful API has no role when securing Backend Application. (2 marks)

Answer: **True** **False**

- 3 Understanding Unit Testing is necessary when testing Backend Application. (2 marks)

Answer: **True** **False**

- 4 Integration Testing is a key concept when managing Backend Application. (2 marks)

Answer: **True** **False**

SECTION B: MULTIPLE CHOICE

Choose ONE correct answer and circle the letter.

- 5 What is the significance of MongoDB when developing RESTFUL APIs with Node JS? (2 marks)

- A. Middleware
B. MongoDB
C. Rate Limiting
D. HTTP status code

Answer: _____

- 6 Which approach to REST API is most effective when developing RESTFUL APIs with Node JS? (2 marks)

- A. Authentication
B. JSON Web Token
C. REST API
D. IIS (Internet Information Services)

Answer: _____

- 7 Which approach to Node.js is most effective when securing Backend Application? (2 marks)

- A. Node.js
B. Request body
C. Authentication middleware
D. Postman testing tool

Answer: _____

- 8 What is the significance of Middleware when securing Backend Application? (2 marks)

- A. Node.js Package Manager

- B. SSL certificate
 - C. Middleware
 - D. Passport.js
- Answer: _____

9 Which approach to REST API is most effective when testing Backend Application? (2 marks)

- A. REST API
 - B. SSL certificate
 - C. Node.js application deployment
 - D. HTTP (Hypertext Transfer Protocol)
- Answer: _____

1 What is the significance of Chai assertion when testing Backend Application? (2 marks)

- A. Middleware
 - B. JSON Web Token
 - C. Cypress
 - D. Chai assertion
- Answer: _____

1 Which approach to RESTful API is most effective when managing Backend Application? (2 marks)

- A. Integration Testing
 - B. RESTful API
 - C. Usability testing
 - D. Hashing algorithm
- Answer: _____

1 What is the significance of Unit Testing when managing Backend Application? (2 marks)

- A. Unit Testing
 - B. Authentication
 - C. Nightwatch.js
 - D. Passport.js
- Answer: _____

SECTION D: OPEN-ENDED

Answer ALL questions. Write in the spaces provided.

1 Analyze the importance of Node.js in developing RESTFUL APIs with Node JS. Identify (13 marks)
3 its advantages and limitations.

1 Evaluate the effectiveness of RESTful API when developing RESTFUL APIs with Node (9 marks)
4 JS. Justify your answer with examples.

- 1** Demonstrate how to use JWT authentication when securing Backend Application. **(9 marks)**
5 Provide a step-by-step explanation.

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- 1** Evaluate the effectiveness of REST API when securing Backend Application. Justify your **(9 marks)**
6 answer with examples.

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- 1** Evaluate the effectiveness of Node.js when testing Backend Application. Justify your **(9 marks)**
7 answer with examples.

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- 1** Analyze the importance of Supertest in testing Backend Application. Identify its **(9 marks)**
8 advantages and limitations.

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- 1** Evaluate the effectiveness of Node.js when managing Backend Application. Justify your **(9 marks)**
9 answer with examples.

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- 2** Analyze the importance of REST API in managing Backend Application. Identify its **(9 marks)**
0 advantages and limitations.

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